

Numerical Linear Algebra First-Year Exam, May 2023

Do 4 (four) problems.

1. This problem has two parts.

- (a) If P is a projector, prove that $\text{null}(P) \cap \text{range}(P) = \{0\}$ and $\text{null}(P) = \text{range}(I - P)$.
- (b) Prove that P is an orthogonal projector if and only if it is Hermitian.

2. Let $\|\cdot\|$ be a subordinate (induced) matrix norm.

- (a) If E is $n \times n$ with $\|E\| < 1$, then show $I + E$ is nonsingular and

$$\|(I + E)^{-1}\| \leq \frac{1}{1 - \|E\|}.$$

- (b) If A is $n \times n$ invertible and E is $n \times n$ with $\|A^{-1}\|\|E\| < 1$, then show $A + E$ is nonsingular and

$$\|(A + E)^{-1}\| \leq \frac{\|A^{-1}\|}{1 - \|A^{-1}\|\|E\|}.$$

3. Define the matrices A and B by

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \\ 2 & 0 & 0 \end{pmatrix}.$$

- (a) Find a singular value decomposition of A .
- (b) Find a QR decomposition of B .

4. Assume $A \in \mathbb{R}^{m \times m}$.

- (a) Prove that $\langle x, y \rangle_A = x^* A y$ is an inner product on $\mathbb{R}^m$ if and only if A is symmetric and positive definite.
- (b) Assume that A is symmetric and positive definite. If x_* is the solution to $Ax = b$ and $\{p_1, \dots, p_m\}$ is an orthonormal basis for $\mathbb{R}^m$ with respect to $\langle \cdot, \cdot \rangle_A$ and $x_* = \sum c_i p_i$, give a formula for the c_i .

5. Let matrix $A \in \mathbb{C}^{m \times n}$ with $n < m$. Let a vector $b \in \mathbb{C}^m$, and let r denote the residual vector $r = b - Ax$.

- (a) Show that x solves the least-squares problem $\min \|b - Ax\|_2$ if and only if $r \in \text{Null}(A^*)$.
- (b) Suppose A is full rank, and describe how to find the least-squares solution using the QR decomposition of A .